Dynamic Behavior study of a synchronous generator Connected to a grid

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***Abstract*—This paper deals with the dynamic modelling and simulation of the isochronous and droop control mode of a gas turbine generator (GTG), connected to an external electrical grid and loads. In this study, the Rowen gas turbine model is used to investigate the dynamic analysis of a GTG system by incorporating of new control techniques, in order to study the behavior of the GTG in island or connected to electrical grid and loads. These techniques consist namely of a Fuzzy-PID controller integrated in the speed loop of the turbine governor control system, an exciter system to control the generated voltages by a synchronous generator as well as a synchronization system. This synchronization technique is designed to overcome the problems of understanding related to the synchronization process of a generator coupled to a running electrical grid. The controlled parameters are namely the turbine speed, the powers and the voltages that are generated by the whole system of GTG-grid. This study is conducted with the MATLAB-Simulink environment, by using data owned from the power plant of the Algiers refinery (SONATRACH) as input parameters. The simulation results show the dynamic response of the system in terms of rise time, overshoot and the duration of the transient state under the different operating modes which we will be used to optimize the protection parameters in the control of the electrical network.**

Keywords— Gas turbine generator; Isochronous/Droop control mode; Fuzzy-PID governor controller; Synchronization system

I. INTRODUCTION

Today, gas turbines are one of the major parts of modern industry. They play a key role in the power generation and mainly in the driving of the synchronous generator [1]. Power generation plants, which are usually interconnected with electrical grid, have very complicated structures, causing problems associated with both the power and the voltage oscillations that are caused by the different connected loads, in particularly the high power induction motors [2].

The electrical synchronous generators may become unstable under arising from faults or from load fluctuations [1], affecting thus seriously the power plant operations and may even lead to a blackout [3]. Hence, the operating points and the settings of the electrical network have to be optimised for the maintaining of the stability and for the continuous functionality. The performances and optimization of electrical installations are more and more required. Always, we search to size the machines, to limit transient voltage drops and to minimize short-circuit currents to relieve the circuit breakers devices. Moreover, it is necessary to know the dynamic response of the machines in terms of the time response generated in speed or power, the resulting overshoots and the duration of the transient regime. This dynamic response can be used to optimize again the protection parameters setting of the digital relays as the overspeed, the overloads and the desynchronization, which will be studied in the future works.

The solutions based on analytical computations are difficult and to slow but today, with the arrival of the computer science and the simulation software have made it easy and faster. Indeed, this allows us to develop models that can reproduce the temporal behaviour of an electrical network with variable topology for various regimes such as the coupling of the generator to the grid and the sharing of the power between the generators. These latest will be discussed in this study. Moreover, these models lets us also to test the control algorithms and new control techniques. A recent published works gives the impact of the computer science on the industrial electrical network field [4]. gives an overview on the automation, the protection and the monitoring of an industrial electrical network. The consequences of faulty synchronization of the generator to the grid [5]. This are inspired us to develop, in this study, a synchronization system that can be useful for both the academic filed and the industrial operators in order to simulate the synchronization process and the operating of the turbo alternator.

Several researches have already studied the control of a gas turbines [6]. Therefore, the aim of this paper is to conceive a GTG simulation model in associating their ancillaries namely, the automatic voltage regulator AVR [7], the power system stabiliser PSS [7], the electrical loads and the synchronous generator [8], in order to study the performances of the whole

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system and to analyse its behavior under the loads variation and the synchronization to the electrical grid.

This study deals with the dynamic modelling and simulation, using Matlab/Simulink. A single shaft gas turbine, modelled by the Rowen technic [9], including a combined Fuzzy-PID controller, in the turbine speed loop under the normal operation conditions [10 - 11]. This allows to perform the isochronous and droop control modes of the GTG system connected to the electrical grid and loads, while observing their behaviour relatively to the disturbances and to the control systems proposed. This GTG system concerns of the Algerian national grid, the synchronous generator, which is driven by the 31.5 MW gas turbine (GE) and supplying different electrical substations (considered as electrical loads). The automatic switch blocks (ref_speed, ref_voltage) figure 7, incorporating both isochronous and droop control mode used to select the references set points in voltage and in power for the AVR and the governor control system. The permutation of the control modes are related to the synchronization system that can be observed from the start-up, until the connecting of the generator to the electrical grid [12]. During the start-up, the isochronous control mode is applied. It controls the turbine speed tills the synchronizing parameters are reached. Then, it switches to droop control mode where the turbine converges towards the set power point. Generally, the droop is 4% and it depends on the turbine speed vs. power characteristic and this to ensure proper load sharing with more than one distributed generators and grid [13].

According to the GTG system and through the simulation, different operating sequences are reproduced, such as start-up of the turbine generator and its synchronization to the electrical grid. The dynamic study is based on the methodology, which consist of supplying consecutively different loads and this by activating their corresponding circuit breakers. These disturbances are taking into account in order to predict the impact of the load variations on the stability of the GTG system alone or connected to the electrical grid. Moreover, the impact of the loads on the GTG-grid system is also evaluated by studying the transient and steady states. The simulation results concern the synchronous speed, the powers, the voltages and the frequencies. The control techniques, the simulation methodology and the results are validated by comparing them to other published works [14 - 16]. the current density injected against the corrosion and the protection of the pipeline coatings.

II. GAS TURBINE MODEL

In 1983, Rowen [10] has developed a simplified model of a gas turbine which may be considered as a reference and reconsidered by many other researchers in their investigations [17]- [7]. It consists of a set of algebraic equations describing the steady state characteristics of a gas turbine [18]. During the normal operating conditions, the frequency and the power are the two fundamental variables controlled by the main control loop [19]. When the operation of the system changes, or becomes abnormal, the temperature controller and the acceleration controller are activated. Generally, during the start-up of the gas turbine the acceleration controller is often used primarily to limit the rate of the rotor acceleration [20].

The turbine torque according the speed and the fuel flow is calculated using the following equation:

$$f_1 = 1.3\left(W_f - 0.23\right) + \frac{1}{2}(1 - N) \qquad (1)$$

However, under normal operating conditions, the simplified Rowen model is illustrated by the following figure.

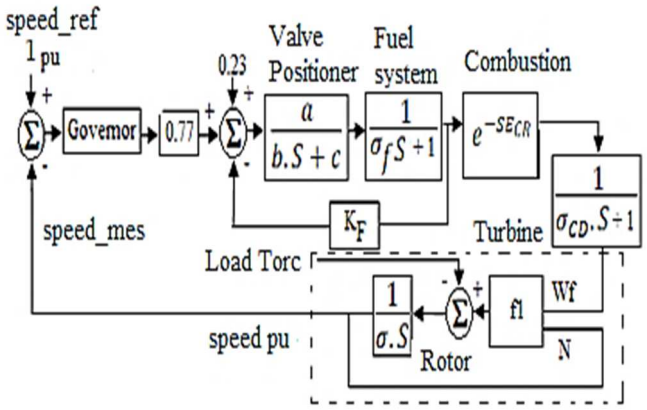


Fig.1. Simplified Rowen transfer function model.

Where W_f and N respectively are the fuel flow and turbine speed in (pu).

The rotor time constant of a gas turbine power plant varies within a wide range from 12.2 s to 25.2 s as data provided by Rowen. So, there is a need to optimize the rotor time constant along with the droop setting value. The equation of the setting time is written with both the droop setting (S) and the rotor time constant (τ) as variables.

$$T_{op} = 14.7983 + 0.155\ s^2 + 0.014.\tau^2 + 0.004.\tau s \qquad (2)$$

III. GOVERNOR SPEED AND POWER CONTROL

Droop and isochronous control mode are the two fundamental methods to control the GTG system in the speed or in the power. In general, the gas turbine drives a synchronous generator with an important part of the power produced, is converted into the electrical power. When the GTG system, or the synchronous generator is connected to the grid, it can be operated in terminal voltage or (Var/power) factor control mode [17].

A very rudimentary explanation of the droop control mode. Is that an increase in megawatt power loading results in a linear decrease in speed, corresponding to the percent droop selected. When the word “isochronous” is applied to the turbine governor speed control, it means that changes in turbine/generator megawatt power loading result in no final speed change from a set reference; to accomplish this as load increases or decreases, immediate speed corrections via the fuel valve are made [21]. Droop control is a commonly implemented independent governor speed control method because it achieves stable electrical power system operation with multiple islanding

system generators [22] and it is used for controlling parallel uninterruptible power supply system's [21].

Many papers are published in the design and setting of the PID and the Fuzzy controllers [23] - [24]. In this paper an optimized controller is proposed, that it is the Fuzzy-PID controller. This combination allows us to design of a new control technique, figure3. In depending of the error and deviation error (e, de), the Fuzzy controller readjusts and modifies the coefficients (kp, ki and kd) of the PID controller in order to improve their static and the dynamic performances.

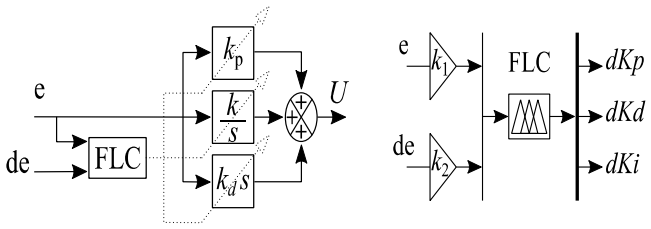


Fig.2. Fuzzy-PID and FLC governor controller

The parameters of the optimized Fuzzy-PID controller are calculated using the following equations.

$$Kp^* = Kp + dKp \quad (3)$$

$$Kd^* = Kd + dKd \quad (4)$$

$$Ki^* = Ki + dKi \quad (5)$$

$$e = \dot{N}_{ref} - \dot{N} \quad (6)$$

The output of the fuzzy-PID algorithm are as follows:

$$U(t) = Kd^*\left(\dot{N}_{ref} - \dot{N}\right) + Kp^*\left(N_{ref} - N\right) + Ki^* \int \left(N_{ref} - N\right)dt \quad (7)$$

$$U(t) = \dot{e}^*(Kd + dKd) + e(Kp + dKp) + Ki + dKi \int e\,dt \quad (8)$$

dKp, dKi *and* dKd are generated and determined by a set of fuzzy rules according to the error and the deviation of error of the speed set point.

$Kp, Kiand\ Kd$ are the values given to the PID controller calculated using le method [25]. The parameters and the coefficients of the controllers are illustrated in table 1.

TABLE I. CONTROLLER PARAMETERS

Controllers	***Kp***	***Ki***	***Kd***
PID-ZN	45	0.5	7.8
Fuzzy-PID	33.6905	0.4045	8.8249

Three positive and negative triangular fuzzy sets intended to cover the input and the output variables as described in figure 2. The inputs and the outputs functions proposed to scheduler the fuzzy logic controller (FLC) is given in figure 3. The inputs functions are (e, de), and the output functions are (dkp, dki and dkd).

The fuzzy subsets are (NL, NM, NS, ZE, PS, PM, PL) which it means for example (NL is Nominal low). There are 49 rules created and implemented under Simulink MAMDANI control technique in the FLC controller.

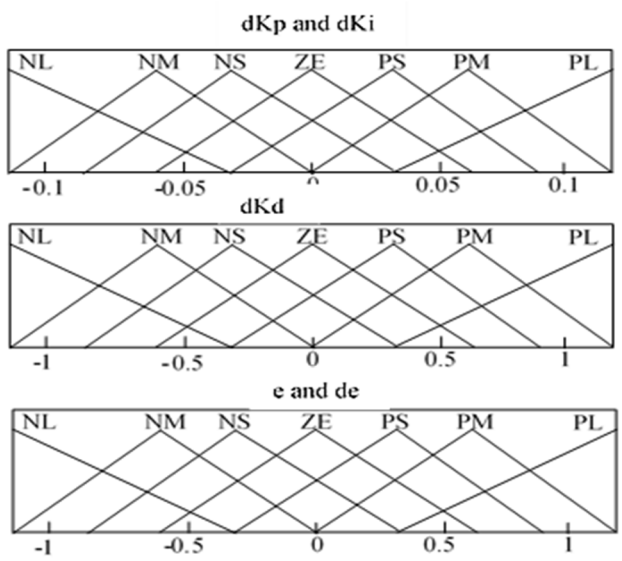


Fig.3. The proposed scheduler functions of the FLC

After the amplification and fuzzification of the input variables (e, de), a correspondence is made between the functions and the subsets of the Fuzzy controller in order to determine the results in linguistic terms of the output variables. Once the fuzzification is verified, it is the role of the defuzzification which drives the process of converting linguistic values into numerical values used to optimize the coefficients of the PID controller [23].

The usual tables characterizing the outputs functions of (dKp, dKi and dKd) of the Fuzzy controller with respecting the inputs (e, de) are given by tables (2,3 and 4).

TABLE II. FUZZY RULE TABLE FOR CHANGE IN DKP

de\ e	NL	NM	NS	ZE	PS	PM	PL
NL	PL	PL	PL	PL	PM	PS	ZE
NM	PL	PL	PL	PL	PM	ZE	ZE
NS	PM	PM	PM	PM	ZE	PS	PS
ZE	PM	PM	PS	ZE	NS	NS	NS
PS	PS	PS	ZE	NM	NM	NM	NM
PM	PS	ZE	NS	NM	NM	NM	NL
PL	ZE	ZE	NM	NM	NM	NL	NL

TABLE III. FUZZY RULE TABLE FOR CHANGE IN DKI

de\ e	NL	NM	NS	ZE	PS	PM	PL
NL	NL	NL	NM	NM	NS	ZE	ZE
NM	NL	NL	NM	NS	NS	ZE	ZE
NS	NL	NM	NS	NS	ZE	PS	PS
ZE	NM	NM	NS	ZE	PS	PM	PM
PS	NM	NS	ZE	PS	PS	PM	PL
PM	ZE	ZE	PS	PM	PM	PL	PL
PL	ZE	ZE	PS	PM	PM	PL	PL

TABLE IV. FUZZY RULE TABLE FOR CHANGE IN DKD

de\ e	NL	NM	NS	ZE	PS	PM	PL
NL	PS	NS	NL	NL	NL	NM	PS
NM	PS	NS	NL	NM	NM	NS	ZE
NS	ZE	NS	NM	NS	NS	NS	ZE
ZE	ZE	NS	NS	NS	NS	NS	ZE
PS	ZE	ZE	ZE	ZE	ZE	ZE	ZE
PM	ZE	PS	PS	PS	NS	PS	PL
PL	PL	PM	PM	PM	NS	PS	PL

The MATLAB/Simulink block of the combined Fuzzy-PID controller is given by figure 4.

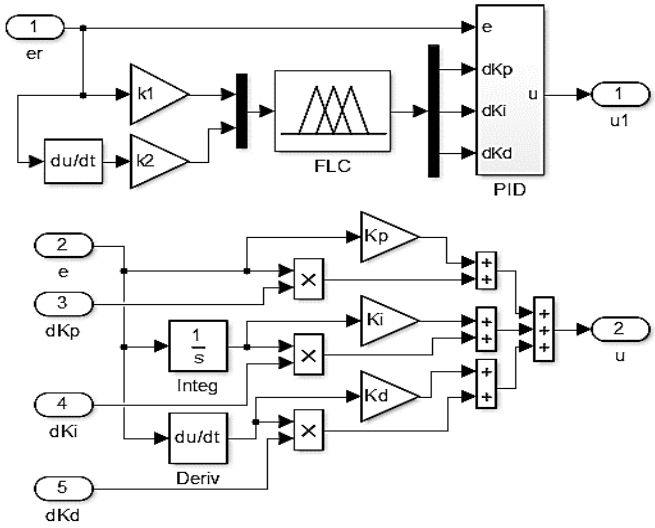


Fig.4. Simulink block of Fuzzy- PID governor controller

The inputs of the error and the deviation of the error are adjusted by the coefficients (k1=5, k2=5) for a rapid processing information.

IV. SYNCHRONIZING SYSTEM

Researchers, engineers, and many other professionals often ask themselves the question: what is the result that I get if I perform an action on an element. Schaefer, are given an overview on the handling of the synchronization system and the power [26].

The objective in this study is not to repeat the requested condition of the synchronization, but to conceive a synchronization system figure 5. In this way, the synchronization technique between the GTG system and the electrical grid can be tested, which it is possible also to simulate the matching process of the turbine speed and the generator frequency to a running electrical grid.

Obviously, we used a phase-locked loop PLL [27], and Park transformation system (abc to dq0) to identify the magnitudes parameters of the voltages, the values of the frequencies and the phases sequences of the synchronous generator and the electrical grid, which these parameters are combined in the state flow controller to perform the synchronization process figure 6.

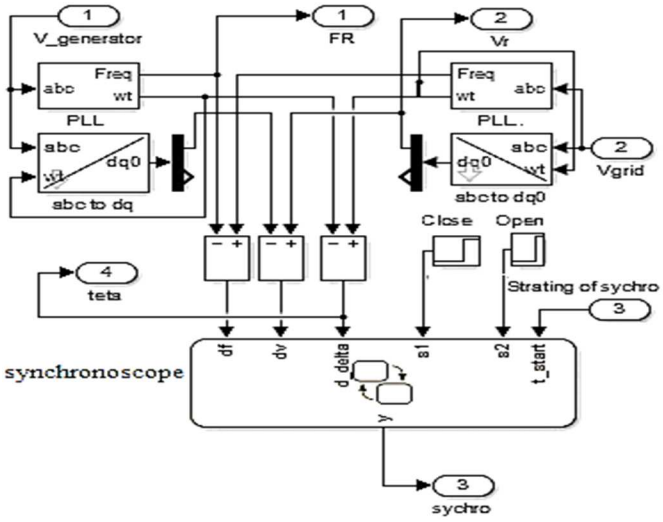


Fig.5. Simulink bloc of the synchronizing system.

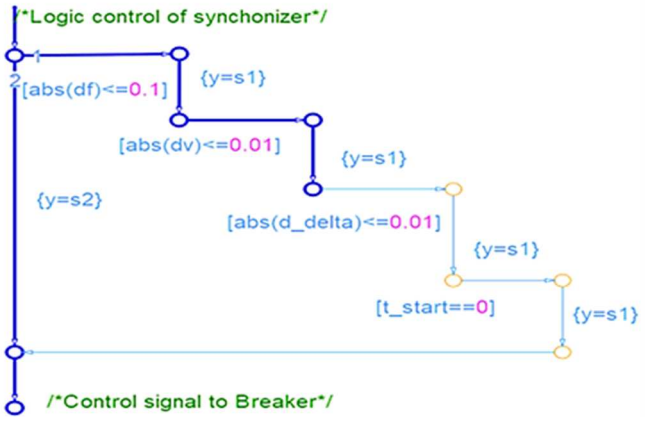


Fig.6. Logic function of the state flow controller according to the synchronization parameters.

This logic of the synchronization system is programmed according to the different absolute deviation errors received by the state flow controller figure 6. These absolute deviation errors of GTG system and grid are successively the deviations absolute errors of the frequencies (df), the deviation magnitude error of the voltages (dv) and the deviation sequences of phases error (d_delta). The signals (s1, s2) are the selected logic functions of the circuit breaker that are alternatively open or

close, and (t_start = 0 or 1) is the permissiveness to initiate the starting of the synchronization. The output signal (y=s1 or s2) is dedicated to control the opening or the closing of the circuit breaker (CB generator) figure 7. The flowchart of the state flow controller is divided in two parallel branches. As described in the figure 6, the primary branch consists of several conditions. If these conditions are verified, the assignment of the controller output (y) takes (y=s1), if not the assignment takes (y=s2). These signals are selected by the synchronisation system to control externally the connecting of the GTG system to the electrical grid.

In general, the synchronization system can be operated manually or automatically [29], and it communicates at the same time with the governor control system (GOV) and the AVR to initiate the startup of the GTG system and to prepare the coupling of the generator to the electrical grid [30].

The isochronous control mode is enabled firstly from the synchronization system, which the GTG system is controlled in speed and in voltage. The voltage set point have to be defined according to the operating state as well as to define the synchronization to an electrical grid. In this case the output signal (y=s2) and the (CB generator) is open. Secondly, and when the synchronizing parameters are reached the output signal of the state flow controller is affected to (y=s1), momentary the (CB generator) is closed and the droop control mode, which the GTG system is controlled in power and the set points of the GOV and the AVR are affected successively to the frequency and the voltages of the electrical grid. In general, the starting of (GE) turbine is achieved by a driven engine up to 60% of the rated speed. After that, the turbine system takes over to reach the rated speed. The exciter system of the synchronous generator is made with permanent magnetic generator (PMG) technology. This latest is assembled with the AVR system, activated after the reaching of 90 % of the nominal speed in order to generate the voltages in generator.

In order to study the operating of the GTG and their behaviour in relation to the supply system of the loads and the sharing of power between the generator and the electrical grid, a simulation methodology is applied to the studied system.

Through the digital simulation, we have reproduced the start-up of the turbine up to synchronization figure 7. Firstly, the loads L1, and L5 are supplied by the electrical grid (SONELGAZ) with the circuit breaker (CB Grid) is closed.

The variables (Vg, Vr); (Pm, Pe, Pr); (fr, fg) are successively the voltages of the generator and the electrical grid; the powers of the turbine, the synchronous generator and the electrical grid; finally, the frequencies of the electrical grid and the synchronous generator. In accordance with figure.7, the Simulink blocs controlled by the state flow synchronizer system are as the following:

- Simulink bloc of (ref_speed)
- Simulink bloc of (ref_voltage)
- Simulink bloc figure 8 (Governor system integrated in Rowen Gas Turbine Model)
- Simulink bloc (CB generator)

At the beginning of the simulation the isochronous control mode is enabled automatically by the synchronoscope. Once the synchronization parameters are verified, the synchronous generator is coupled to the electrical grid, at the same time the droop control mode is enabled, and the turbine converges to the set power point configured at 0.2 pu firstly.

Secondly, the remaining loads L2, L3 and L4 are successively activated by closing their own circuit breakers at different times 20s, 30s and 60s in order to study the impact of the variation of the loads on the stability of the GTG-grid system and during the sharing of the power between the GTG system and the turbine electrical grid.

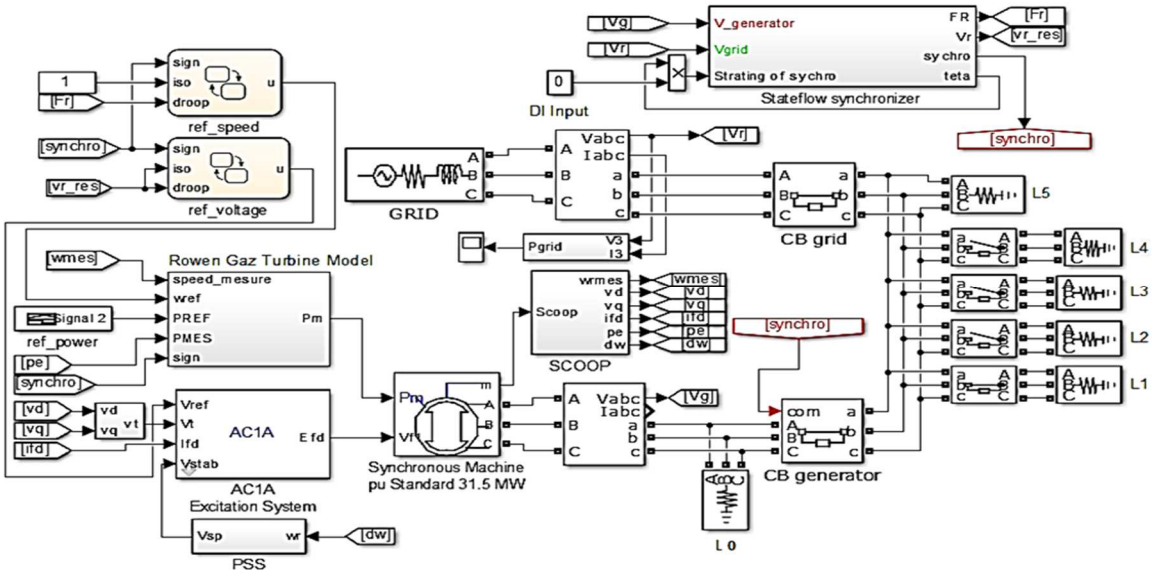


Fig.7. Simulink bloc of the GTG system with their elements constituted of the synchronizer system, the governor control turbine, the exciter AVR, power system stabilizer PSS, electrical loads and grid

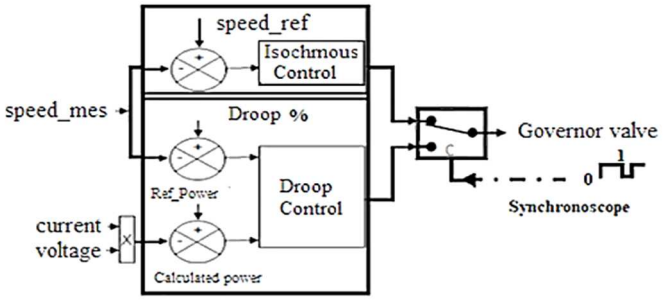


Fig.8. Communication and functional block of the governor control system in the GTG system.

V. RESULTS AND ANALYSIS

The simulation has been achieved according to the procedures described above in order to study the behavior of the GTG system and to demonstrate the operating of a turbo-generator in island, and when it is connected to the electrical grid. Electrical loads are connected to the different circuit breakers activated and deactivated according to the prescribed times in the aim to simulate the approach actual operation of the GTG system which we can show how the power is shared between the synchronous generator and the electrical grid, the stability of the GTG- grid system and the performances of control techniques proposed.

The operating process and the different control modes of the GTG system under the effect of the disturbances are reproduced such as, the start-up of the GTG system, the synchronization of the synchronous generator to the grid, which the transient state and the steady state are observed during these different control modes.

The dynamic responses of the GTG system are displayed by the different characteristics namely by the turbine speed, the powers generated, the voltages, the frequencies presented as a function of the simulation time by figures 9 to 18 where they are compared to others results found in other similar published papers.

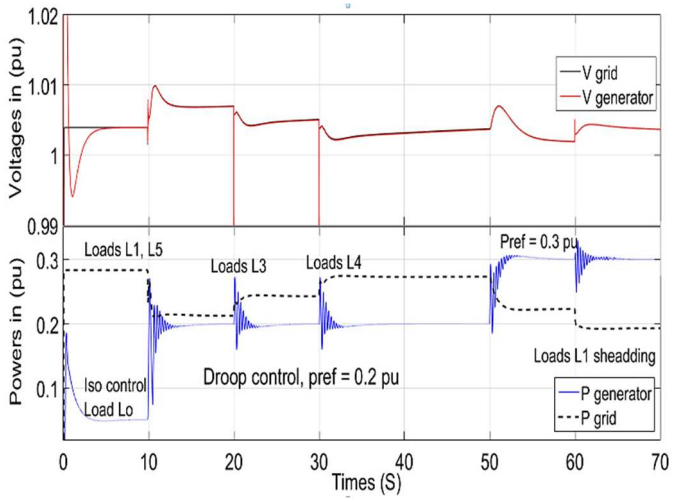


Fig.9. Powers and voltages generated by (GTG and Grid).

According to the figure 9, the isochronous control mode lasts for about (t=9,8 s) before the gas turbine generator is synchronized to the electrical grid. At the time of (t=10.2 s) the generator circuit breaker (CB generator) is closed from the synchronization system, the generated voltage of the synchronous generator and the electrical grid voltage coincides as shown.

The droop control mode is then enabled automatically and the turbine generator converges to the set power point, that it is, 0.2 pu. This previous power generated is maintained constant despite the load's variation (L3, L4) with a power of 5MW for each load activated at different times, t=20 s, 30 s. The increasing of the set power point to 0.3 pu at 50 s and the shedding of the load (L1 = 5MW) at 60 s. keeps also the operating around the programing setpoint. It should be noted that any increase in the set power point of the turbine from 0.2 to 0.3 pu leads to a decrease in the grid power as seen at t=50s, and the variation of loads generates increases as well as the decreases of the power generated by the electrical grid and some fluctuations are recorded in the power generated by the generator temporarily which stabilizes later.

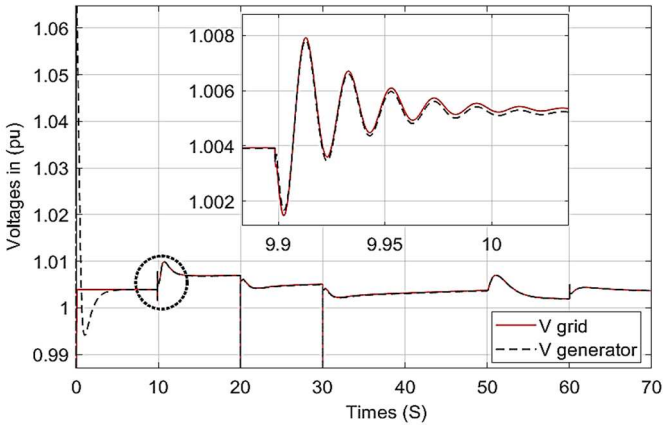


Fig.10. Zoom of the voltages magnitude during the coupling of the generator to electrical grid.

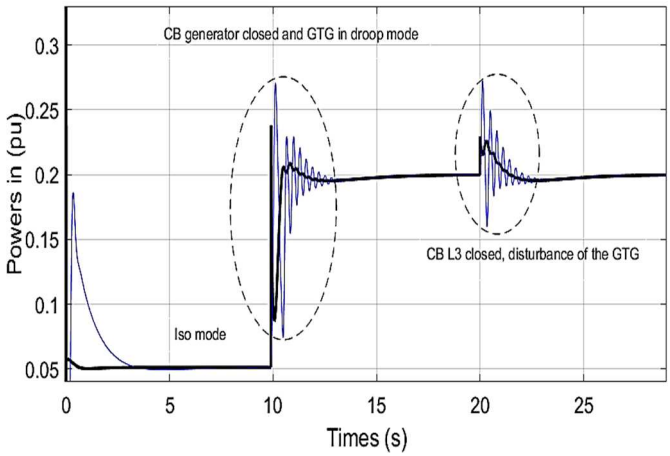


Fig.11. Mechanical and electrical power generated by the turbine and generator during the coupling to the grid and as a result of the load (L3) disruption.

From the figures 12, it can be seen that the mechanical power developed by the turbine and the generated speed are very smooth with small fluctuations, which leads to conclude about the effectiveness of the optimized Fuzzy-PID governor controller.

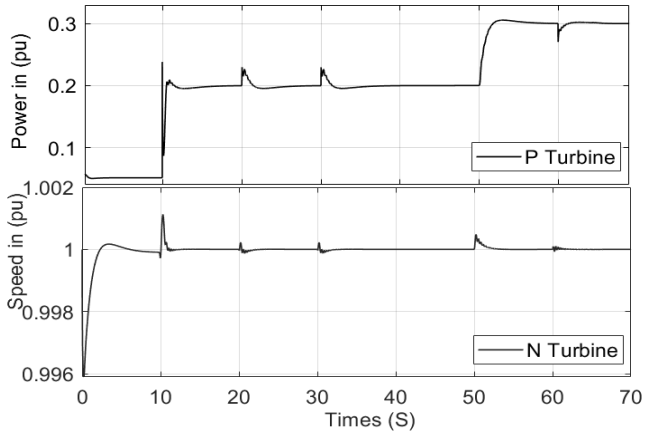


Fig.12. Mechanical power and the speed generated by the turbine during the disturbances.

Moreover, the generated power and the voltages follows perfectly successively the set power point and the set point of the electrical grid voltage after each transient period figure 10, 11 and 12. This may lead to conclude the good functioning of the Simulink studied model which it kept stable the operating of the GTG system during all the different disturbances.

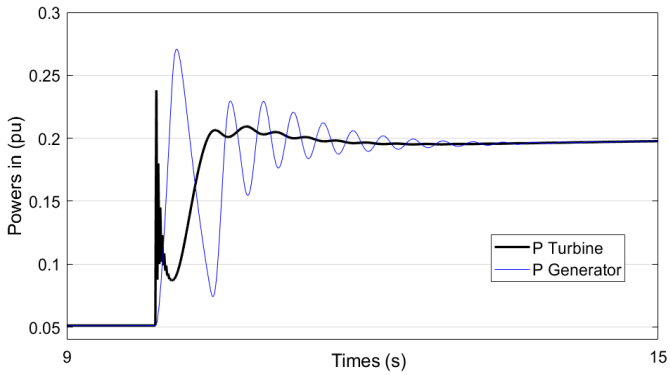


Fig.13. Superposition zooms of the mechanical and electrical powers of the GTG during the coupling to grid.

Furthermore, the figures 11 and 13 characterizes the superposition of the electrical power and the mechanical power generated by the GTG system during the coupling of the synchronous generator to the electrical grid. It can be seen that the steady state is reached after a very short period of time (about 3 s). The overshoot of these generated powers is considerable, passing from a minimum power value to an upper value.

The dynamic analysis of the GTG system responses such as the characteristics of the transient state and the steady state of the turbine speed, the mechanical and the electrical powers generated, the voltages and the frequencies during the synchronization are taken into a consideration in order to verify and to validate this research work. The observed dynamic responses of the GTG system synthesis the performances of the control techniques in terms of the time response, the overshoot and the transient state according the simulation methodology and the different control modes applied.

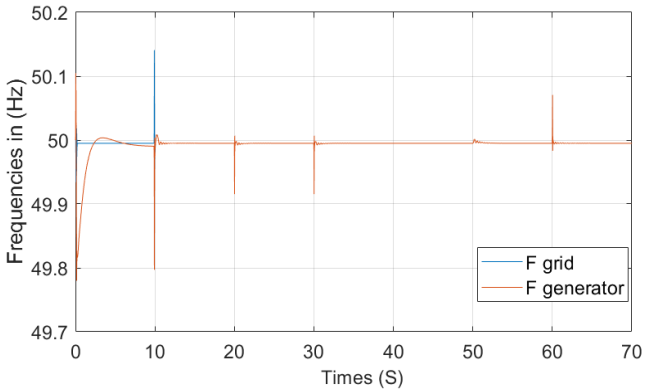


Fig.14. Superposition of the frequencies generated by the system (GTG-grid) during the isochronous and droop control mode and the sharing powers between the synchronous generator to the electrical grid.

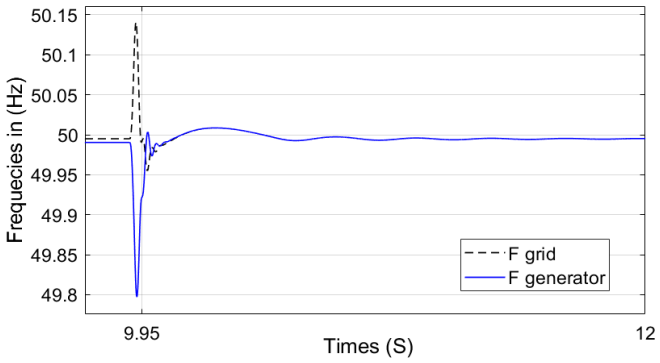


Fig.15. Frequencies superposition zoom during the synchronization of the GTG to grid.

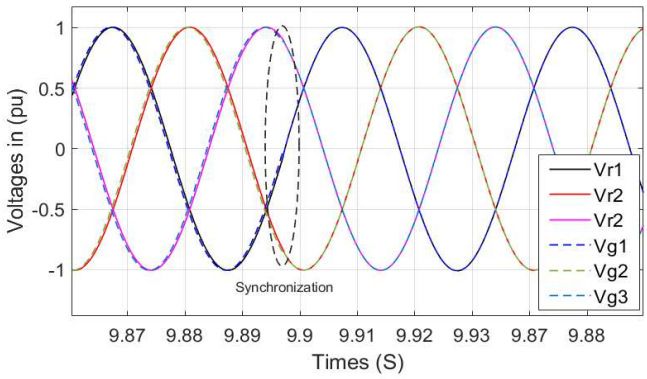


Fig.16. Voltages superposition during the synchronization of the GTG to grid.

From the conducted simulations the behavior of the GTG system is studied at the around essentially of the operating parameters of the frequency and the voltages (50 Hz, 11 kV). Often this research type is carried out purely in the industries.

Moreover, we have opted to compare the obtained results to others similar results published papers in the same domain application which deals the synchronization and the sharing

powers between the generators and grid. The figures 17 and 18 describe consecutively an overview on the control of the power and the effect of the variation of the set power point on the generated frequency, as well as the allures of the transient frequencies during the paralleling of the generators.

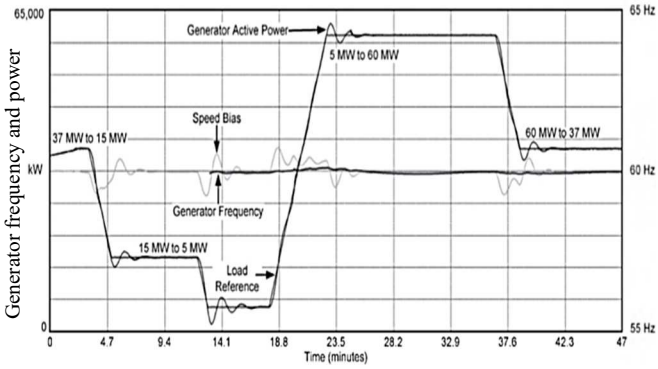


Fig.17. Dynamic characteristics of the active power and frequency generated by a turbogenerator during the variation of the set power point [17].

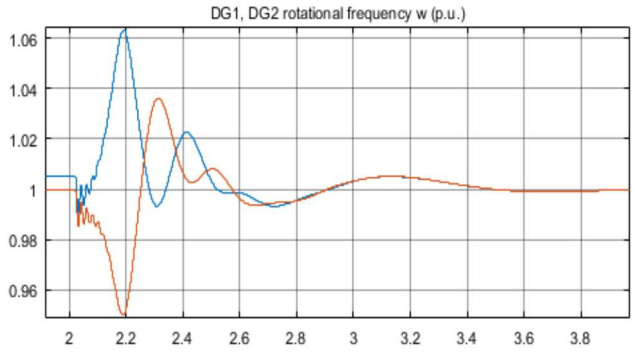


Fig.18. Dynamic characteristics of the transient frequencies during paralleling of two generators [19].

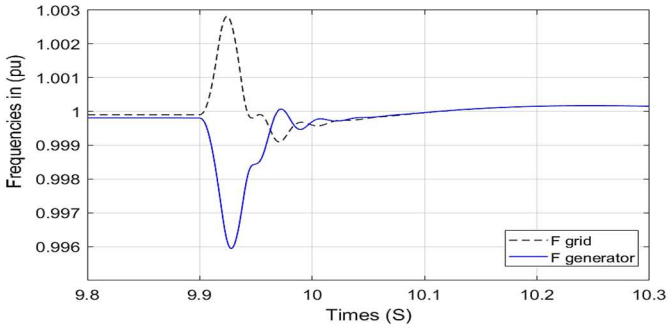


Fig.19. Dynamic characteristics of the transient frequencies during paralleling of the GTG to grid.

The results obtained show clearly the behavior and the functioning of the GTG system in island and connected to the electrical, which we can see the good performances of Fuzzy-PID governor controller in terms of time response and steady state error during the control of the speed and power, figure 12.

From the figures 15 and 16, a very small frequencies and voltages overshoot recorded during the synchronization of the synchronous generator to the grid and during the disturbances which we can validate the proposed control techniques of the synchronization system and the communication with the governor and the AVR control systems.

VI. CONCLUSION

This study is focused on the control strategies of a GTG system connected to the electrical grid and loads. This GTG system operates in different modes when a change in the state is requested. It operates in droop control mode while the system is connected to the electrical grid and in isochronous mode while the system is in island. The exploring reaction of the GTG system to the disturbances is still a challenging issue. Therefore, there is an increasing demand for accurate dynamic models to investigate the dynamic response to the disturbances and to improve the existing control systems. Thereby, this GTG system is combined of several models namely, the gas turbine generator, the PSS, the AVR and the control systems. These latest, are designed and simulated under MATLAB/Simulink environment. The Simulink model can cover research activities for different kinds of GTG system, including low power, industrial power plant generator and the control systems.

The simulation results show the operating and the behaviour of the GTG system during the occurring of a disturbance at which the dynamic responses, in term of the turbine speed and the generated powers with their time response, the overshoot resulting and the duration of the transient regime, are determined. These simulation results can be used to conduct other researches projects concerning the optimization of the protection parameters in the digital relays of turbo-generators against the overspeed, the overloads, the short-circuit currents, the shaft line protection, the loss of excitation and the desynchronization. Moreover, the simulation results display the characteristics of the turbine speed, the powers, the voltages and the frequencies under the influence of the variation of the loads.

Otherwise, these results are compared to other published theoretical and experimental researches and it shows a good concordance [17 - 19]. This allows us to validate the proposed Simulink model, the simulation methodology and the performances of the control systems, especially the fuzzy-PID controller and the synchronisation system.

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